ANSWER KEY & MARKING SCHEME — SET A

Chapter 1: **Chemical Reactions and Equations** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** MgO	$2\,Mg + O_2 \longrightarrow 2\,MgO$
2	**(d)** combination and exothermic	Two substances combine; heat is released
3	**(c)** 3, 4, 1, 4	$3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$
4	**(c)** Incorrect — light or electricity can also be used	e.g. $2\,AgCl \xrightarrow{\text{light}} 2\,Ag + Cl_2$
5	**(a)** Student A is correct	Zn displaces Cu
6	**(b)** decomposes into silver and chlorine	Photolytic decomposition
7	**(d)** double displacement (precipitation)	Exchange of ions; white $BaSO_4$ precipitate
8	**(a)** H_2	H_2 gains oxygen
9	**(c)** nitrogen	Inert, prevents oxidation of fats
10	**(a)** 2 : 1	$2\,H_2O \longrightarrow 2\,H_2 + O_2$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It is covered with a layer of magnesium oxide/basic magnesium carbonate, which prevents it from burning; cleaning removes this layer.	1+1
12	$H_2 + Cl_2 \longrightarrow 2\,HCl$ — combination reaction	1+1
13	Slow eating away of a metal by air, moisture or chemicals around it; e.g. rusting of iron (also black coating on silver, green coating on copper).	1+1
14	False. In respiration glucose is oxidised: $C_6H_{12}O_6 + 6\,O_2 \longrightarrow 6\,CO_2 + 6\,H_2O$ + energy — it releases energy, so it is exothermic.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) $3\,H_2 + N_2 \longrightarrow 2\,NH_3$ (ii) $2\,H_2S + 3\,O_2 \longrightarrow 2\,H_2O + 2\,SO_2$ (iii) $3\,BaCl_2 + Al_2(SO_4)_3 \longrightarrow 3\,BaSO_4 + 2\,AlCl_3$	1+1+1
16	Green crystals turn white then reddish-brown; smell of burning sulphur. $2\,FeSO_4 \xrightarrow{\text{heat}} Fe_2O_3 + SO_2 + SO_3$ — thermal decomposition	1+1+1
17	Brown coating on the nail; blue solution turns light green. $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ — displacement	1+1+1
	OR: Oxidation: gain of oxygen/loss of hydrogen, e.g. $2\,Cu + O_2 \longrightarrow 2\,CuO$. Reduction: loss of oxygen/gain of hydrogen, e.g. $CuO + H_2 \longrightarrow Cu + H_2O$ (CuO reduced).	1½+1½
18	(i) Brown fumes of NO_2 (ii) $2\,Pb(NO_3)_2 \xrightarrow{\text{heat}} 2\,PbO + 4\,NO_2 + O_2$ (iii) thermal decomposition	1+1+1
	OR: Paint prevents contact of iron with air and moisture, so it does not rust. Nitrogen replaces oxygen and prevents oxidation of fats/oils (rancidity).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Equal number of atoms of each element on both sides; law of conservation of mass (2). (a) $2\,Na + 2\,H_2O \longrightarrow 2\,NaOH + H_2$ — displacement (b) $Pb(NO_3)_2 + 2\,KI \longrightarrow PbI_2 + 2\,KNO_3$ — double displacement (c) $2\,Al + 3\,CuCl_2 \longrightarrow 2\,AlCl_3 + 3\,Cu$ — displacement (3)	2+3
	OR: (i) $CaCO_3 \longrightarrow CaO + CO_2$ (ii) $2\,AgCl \longrightarrow 2\,Ag + Cl_2$ (iii) $2\,H_2O \longrightarrow 2\,H_2 + O_2$ (3). AgBr decomposes in light: $2\,AgBr \longrightarrow 2\,Ag + Br_2$ (2)	3+2
20	(i) Hydrogen (ii) burns with a pop sound (iii) $Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2$ (iv) displacement (v) no reaction — Cu is less reactive than H	1+1+1+1+1
	OR: (i) Oxidation of fats/oils making food smell and taste bad (ii) antioxidants, nitrogen flushing, airtight containers, refrigeration (iii) hydrated iron(III) oxide, $Fe_2O_3 \cdot x\,H_2O$ (iv) painting, oiling/greasing, galvanising	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 1: **Chemical Reactions and Equations** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** melting of ice	No new substance is formed
2	**(d)** combination reaction	Two reactants form one product
3	**(d)** exothermic	Heat is released
4	**(c)** Incorrect — the number of atoms of each element must be equal	e.g. $2\,H_2 + O_2 \longrightarrow 2\,H_2O$: 3 molecules → 2
5	**(a)** Student A is correct	One reactant breaks into two products
6	**(c)** NO_2	Nitrogen dioxide
7	**(b)** silver chloride	$AgNO_3 + NaCl \longrightarrow AgCl\downarrow + NaNO_3$
8	**(d)** gain of oxygen or loss of hydrogen	Definition
9	**(a)** iron displaces copper, forming $FeSO_4$	Fe is more reactive
10	**(c)** CuO	$2\,Cu + O_2 \longrightarrow 2\,CuO$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Glucose is oxidised in cells to CO_2 and water, releasing energy: $C_6H_{12}O_6 + 6\,O_2 \longrightarrow 6\,CO_2 + 6\,H_2O$ + energy.	2
12	$Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2\,Fe$ — displacement (redox), highly exothermic	1+1
13	In combination, two or more substances form one; in decomposition, one substance breaks into two or more. e.g. $2\,H_2O \longrightarrow 2\,H_2 + O_2$ is the reverse of $2\,H_2 + O_2 \longrightarrow 2\,H_2O$.	1+1
14	False — it is photolytic decomposition: $2\,AgCl \xrightarrow{\text{light}} 2\,Ag + Cl_2$; the grey colour is due to silver.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) $BaCl_2\,(aq) + Na_2SO_4\,(aq) \longrightarrow BaSO_4\,(s) + 2\,NaCl\,(aq)$ (ii) $NaOH\,(aq) + HCl\,(aq) \longrightarrow NaCl\,(aq) + H_2O\,(l)$ (iii) $Zn\,(s) + 2\,AgNO_3\,(aq) \longrightarrow Zn(NO_3)_2\,(aq) + 2\,Ag\,(s)$	1+1+1
16	(i) Na oxidised, O_2 reduced (ii) H_2 oxidised, CuO reduced (iii) HCl oxidised, MnO_2 reduced	1+1+1
17	Exothermic: heat released, e.g. $CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$. Endothermic: heat absorbed, e.g. $CaCO_3 \xrightarrow{\text{heat}} CaO + CO_2$.	1+1+1
	OR: (i) Calcium hydroxide (slaked lime) (ii) combination, exothermic (iii) $Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$ — a thin shiny layer of $CaCO_3$ forms	1+1+1
18	(i) Hydrogen at cathode, oxygen at anode (ii) 2 : 1 (iii) to increase the conductivity of water	1+1+1
	OR: Silver reacts with sulphur compounds in air forming silver sulphide (black). Copper reacts with moist CO_2 forming basic copper carbonate (green).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) decomposition (b) displacement (c) double displacement (precipitation) (d) combination (e) redox (ZnO reduced, C oxidised)	1+1+1+1+1
	OR: (i) Slow destruction of metal by surroundings (ii) Nails in: A — water + air (rusts); B — boiled water with oil layer (no air, no rust); C — dry air with anhydrous $CaCl_2$ (no rust) (iii) galvanising/painting	1+3+1
20	(i) X = copper, Y = copper(II) oxide (ii) $2\,Cu + O_2 \longrightarrow 2\,CuO$ (iii) combination/oxidation (iv) pass hydrogen over hot CuO: $CuO + H_2 \longrightarrow Cu + H_2O$	1+1+1+2
	OR: (i) Lead nitrate (ii) $2\,Pb(NO_3)_2 \longrightarrow 2\,PbO + 4\,NO_2 + O_2$ (iii) thermal decomposition (iv) nitrogen dioxide (v) lead(II) oxide	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.